

### The Lorentz system

The Lorentz system was introduced by meteorologist Edward N. Lorenz in 1963 as a simplified model of atmospheric convection. While studying the equations numerically, Lorenz found that tiny differences in initial conditions could lead to dramatically different outcomes. This sensitivity became a defining feature of deterministic chaos, often illustrated by the butterfly effect. The Lorenz equations remain a classic example of how simple nonlinear systems can produce complex, unpredictable behavior.

The Lorentz system is written in the form of a general dynamical system

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}),$$

where

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; \quad \mathbf{f}(\mathbf{x}) = \begin{bmatrix} s(y - x) \\ rx - xz - y \\ xy - bz \end{bmatrix},$$

and  $b$ ,  $s$  and  $r$  are suitable constants. Time evolution of this system, can be tracked using the simple forward Euler method

$$\frac{d\mathbf{x}}{dt} \approx \frac{\Delta\mathbf{x}_n}{\Delta t} = \mathbf{f}(\mathbf{x}_n),$$

which results in

$$\mathbf{x}_{(n+1)} = \mathbf{x}_{(n)} + \mathbf{f}(\mathbf{x}_n) \cdot \Delta t.$$

This simple time integration is stable for sufficiently small timesteps  $\Delta t$ . However, depending on the parameters  $(b, s, r)$ , the system shows different dynamics. Observe and report on the following modes.

1. **Periodic solution:** Set  $(s, r, b) = (10, 15, \frac{8}{3})$ .

- Begin from  $(x, y, z) = (0, 1, 0)$ , find the solution for  $t \in [0, 1000]$ .
- Plot the  $x = x(t)$ ,  $y = y(t)$  and  $z = z(t)$ , as well as  $(x, y)(t)$ ,  $(x, z)(t)$ ,  $(y, z)(t)$  and also  $(x, y, z)(t)$ .
- Find the center of the periodic solution (that is the position of attractor).

2. **Chaotic solution.** Set  $(s, r, b) = (10, 50, \frac{8}{3})$ .

- Begin from  $(x, y, z) = (0, 1, 0)$ , find the solution for  $t \in [0, 1000]$ .

- Plot the  $x = x(t), y = y(t)$  and  $z = z(t)$ , as well as  $(x, y)(t), (x, z)(t), (y, z)(t)$  and also  $(x, y, z)(t)$ .
  - Is there any attractor for this solution. Discuss your statement.
3. Repeat the above steps for the initial condition  $(x, y, z) = (0, 1.001, 0)$  and observe the differences. Discuss your results.